

1. Introduction

1.1. Purpose of the system

The system is used for entertainment of the user. It is supposed to be a challenging experience which forces the player to evolve his skills and navigate carefully through the environment of the game.

1.2. Scope of the system

The System is quite small as it is only used for educational purposes and to get used to the processes of developing a video game.

1.3. Objectives and success criteria of the project

The objective of the project is to create a small video game with at least a playable level and a completely planned out and structured development.

1.4. Definitions, acronyms, and abbreviations

The playable character is named "L41k4".

The research facility is named "Heidelberg Dynamics"

2. Proposed system

2.1. Functional requirements

The player is required to be able to move the playable character throughout the levels. WASD for up, down, left, right.

The player character should be able to aim using the mouse, e.g. the character aims in the cursor direction at all times.

The player character should move throughout a level, interacting with the environment. e.g. bump into a wall, walk down a corridor, etc. (Collision).

The playable character and the enemies have a certain number of hitpoints.

Projectiles have a limited range.

Projectiles get destroyed when colliding with another object (i.e. walls, shelves, etc.).

When two projectiles collide they cancel each other out.

When the hitpoints of an entity run out, the character is eliminated.

The health and shield of the entities is limited between two values that are immutable.

The game is required to have a useable weapon which the player can use to reduce the hitpoints of enemies.

There also have to be different enemies.

The enemies can attack and damage the player.

The player must also be able to defeat these enemies with a certain amount of attacks.

There are keycards the player can collect in each level.

To complete a level the player must be able to collect the keycards first.

The game features a health system that manages all the damage and health variables of all entities in the game.

There are first aid kits that replenish the player's health and shield.

There is a enemy type "turret" that aims at the player and shoots projectiles at them.
There is a enemy type "bossturret" that aims at the player and shoots projectiles at them but at a significant faster pace as the normal "turret".
The player can win the game by defeating the two bossturret enemies in the last level.

2.2. Nonfunctional requirements

2.2.1. User interface and human factors

The game features a health and a shield bar that is visible through a HUD to the player at all times.

The game contains a indicator that shows the player if they already collected all keycards of the current level.

After defeat, the player can restart the game with a press of a button.

The program can be closed with the press of a button.

The controls of the game can be learned by the within 3 minutes.

2.2.2. Documentation

There should be a documentation of at least the most important diagrams that help to understand the concept of the game in detail and as a whole. (see 2.5)

There should be a short documentation of all SCRUM meetings.

There should be a documentation of changes made in the development process.

2.2.3. Hardware considerations

A PC with at least Windows 10 or a contemporary GNU/Linux variant,

At least 2 GB RAM

At least a GTX 750 or equivalent GPU

At least 2 GB of SSD Storage

At least a Ryzen 3 1200/Core i5-5200U or equivalent.

2.2.4. Performance characteristics

The game should run with at least 30 FPS.

A reaction to input should happen within 30 ms.

2.2.5. Error handling and extreme conditions

Errors and exceptions should be logged and reported to the developers.

2.2.6. Quality issues

All the code is written in Chromium coding style.

We use line breaks after 80 characters.

We always use Test-Driven Development.

A class has a maximum of 200 LOC.

2.2.7. System modifications

There is just a log used for error handling.

2.2.8. Physical environment

There is no internet connection required to play the game.

There is a mouse and keyboard needed to play the game.

There are no accessibility settings due to the games short development time.

2.2.9. Security issues

Due to the fact that the game uses no internet connection and no data from the player so there should be no security issues.

2.2.10. Audio

The game features its own soundtrack that is running in the background during the playthrough.

There is a sound effect when the player shoots a projectile.

There is a different sound effect when enemies shoot projectiles.

There are sounds when the player is hit.

There are sounds for hitting enemies.

2.3. Pseudo requirements

The game should feel smooth and responsive at all times.

The game is designed in pixelart.

All levels and the overall design of the characters should match the setting of the research facility.